

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Built and shipped features across a multi-tenant NHS eTriage stack (**Node.js, MongoDB**) for **8+ UK clients** that real clinical users depended on, processing **10,000+ monthly submissions**; contributed to **\$340K+** in revenue.
- Architected a full-stack platform end to end (**React, Node.js, MongoDB**) used daily by the AI team, cutting manual work by **100+ hours/month** and lifting throughput by **60%**.
- Wrote clean, production-ready backend code with input validation, integration tests, auth hardening, and AWS KMS encryption for sensitive data, remediating **20+ VAPT findings**.
- Owned the eTriage backend as primary technical POC, shipping features to production in a fast feedback loop, mentoring a second developer, and piloting a COO-sponsored GitHub Copilot experiment through PR review.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs (Node.js, Fastify, PostgreSQL)** with validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core repositories from JavaScript to **TypeScript** and shipped a production LLM auto-commenting feature (**4-bit quantized Llama-7B via llama.cpp**) with async queued generation and content-safety filtering.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and shipped **Underdogs Fitness** (Website), a production **MERN** platform (React, Node.js, MongoDB) with JWT auth, Stripe payments, and cron automation, independently end to end; live **3+ years**.

Projects

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers, Durable Objects, D1

- Built a full-stack app end to end in **TypeScript**: a streaming AI research assistant with a WebSocket client, durable restart-safe workflows using Workers AI for embeddings and inference, and D1 storage, so multi-step tasks survive process restarts and tool failures.

Sonare – Offline Sign / Speech App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Engineered a **React**/Electron frontend and FastAPI backend integrating camera, microphone, and local services (MediaPipe hand tracking, whisper.cpp ASR, local TTS) into one offline application, with async APIs and queues keeping the UI responsive under local video and audio streams.

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python/FastAPI backend service turning open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with cell-level provenance; hardened with **150+ tests** and provider routing, deployed as a live cloud service.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: TypeScript, JavaScript, Python, SQL, MongoDB, PostgreSQL, SQLite

Frontend & Web: React, Next.js, Tailwind CSS, HTML/CSS, Electron, REST APIs

Backend & APIs: Node.js, Express, FastAPI, async services, multi-tenant architecture, integration testing

Tools & Practices: Git, GitHub Actions, CI/CD, Docker, Agile / Scrum, AI-assisted dev (Copilot)

Cloud & Platforms: AWS, GCP, Azure, Linux, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.